CanSat & CubeSat
Ground Control Software (GCS)

Web Based Ground Control Station for Telemetry Monitoring,
Real-Time Visualization, and Mission Control

INDIA SPACE LAB

Project Report

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Student Name : Krupa Devaraj Ramisetty

Institute Name : VIT-AP University

Institute Roll no. : 23BCE9136

Enrollment no. : ISL-391786

Aim

The aim of this project is to design and develop a professional single-page Ground Control Software (GCS) capable of monitoring telemetry data, displaying mission parameters, visualizing GPS location, and simulating mission operations for a CanSat mission.

Objectives

- To understand the concept of Ground Control Software.
- To develop a single-page aerospace dashboard.
- To display telemetry information in real time.
- To visualize mission data using graphs.
- To implement GPS tracking functionality.
- To simulate mission control operations.
- To improve understanding of aerospace telemetry systems.

Introduction

A CanSat is a miniature satellite designed to fit within the dimensions of a standard soft drink can. It is widely used in aerospace education to demonstrate satellite communication, telemetry, navigation, and mission operations. Ground Control Software (GCS) acts as the communication interface between the satellite and the operator by receiving telemetry data, displaying mission parameters, and allowing mission control commands.

In this project, a web-based Ground Control Software dashboard was developed using HTML, CSS, and JavaScript.

The dashboard provides a professional aerospace-style interface with telemetry visualization, interactive graphs, GPS tracking, and mission control buttons.

Theory

Ground Control Software is an essential component of satellite missions. It continuously receives telemetry packets from the satellite and presents important mission parameters such as altitude, temperature, pressure, battery voltage, and GPS location. Operators use this information to monitor satellite health and make mission decisions.

Telemetry refers to the automatic collection and transmission of data from a remote system to a monitoring station. In aerospace applications, telemetry enables engineers to analyze the satellite's condition in real time.

Visualization tools such as graphs and maps improve situational awareness by presenting mission data in an intuitive format. Modern GCS applications also support command execution, fault monitoring, and data export capabilities.

Software Used

- HTML5
- CSS3
- JavaScript
- Chart.js
- Leaflet.js
- OpenStreetMap
- Visual Studio Code

- Google Chrome

System Design

The developed Ground Control Software consists of the following modules:

- Top Control Panel
- Telemetry Display
- Mission Control Buttons
- Real-Time Altitude Graph
- GPS Tracking Map
- Error Code Display
- Battery Monitoring

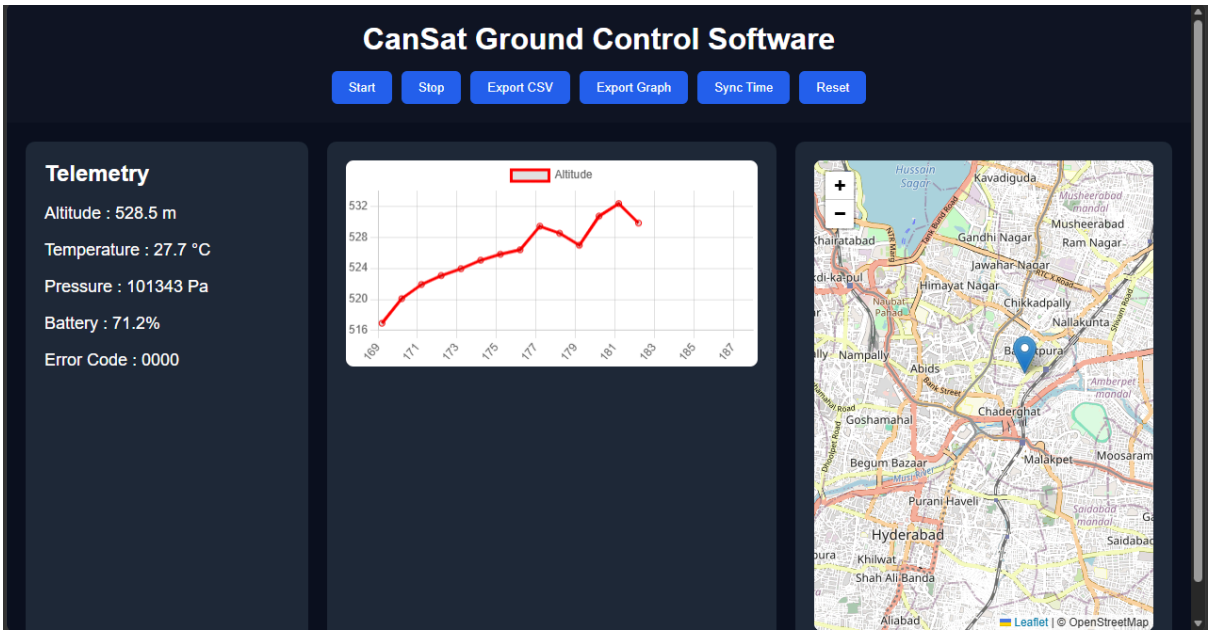
The dashboard was designed with a responsive layout for improved readability and ease of operation.

Implementation

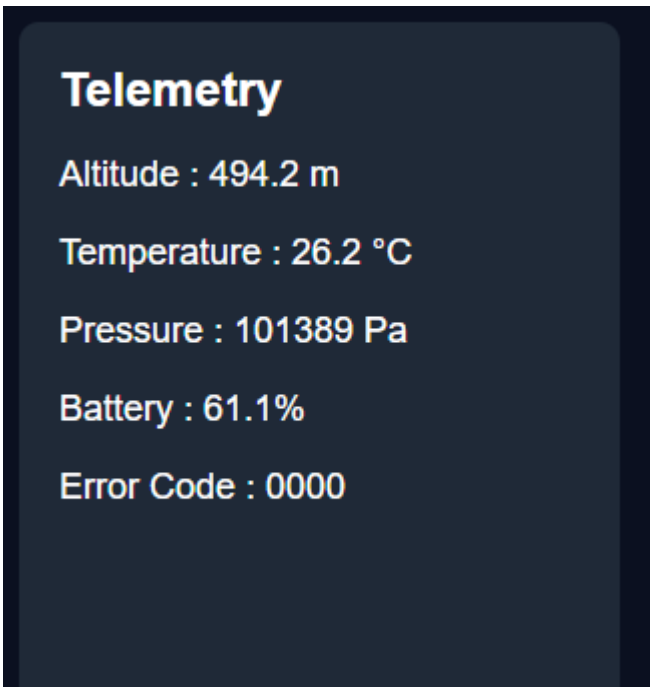
The interface was developed using HTML for structure, CSS for styling, and JavaScript for dynamic functionality. Simulated telemetry values were generated to represent altitude, temperature, pressure, and battery status. A real-time line chart was implemented using Chart.js, while GPS tracking was displayed using Leaflet.js with OpenStreetMap. Mission control buttons were included to simulate telemetry control and mission operations.

Results

- Dashboard Home Screen



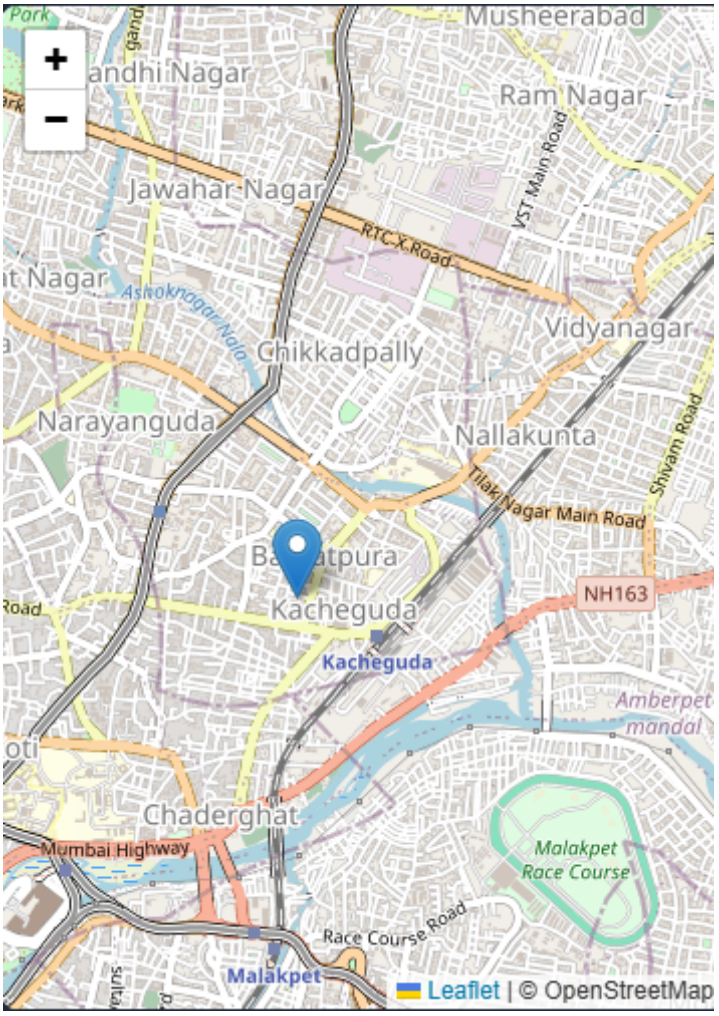
- Telemetry Display



- Live Altitude Graph



- GPS Tracking Map



Observations

- The dashboard successfully displayed simulated telemetry values.
- Real-time altitude graphs updated continuously.
- GPS location was visualized using an interactive map.
- Mission control buttons provided a simple operational interface.
- The overall interface was responsive and user-friendly.

Conclusion

The project successfully demonstrated the development of a web-based Ground Control Software for a CanSat mission. The dashboard integrated telemetry monitoring, graphical visualization, GPS tracking, and mission control features into a single interface. The project provided practical experience in aerospace software development, front-end web technologies, and telemetry visualization concepts.

References

1. India Space Lab – CanSat Ground Control Software Assignment.
2. MDN Web Docs.
3. Chart.js Documentation.
4. Leaflet.js Documentation.
5. OpenStreetMap Documentation.